

Proof of Corollary 5.6

Let $\Lambda := \{A \in \mathcal{P}(X) : \sigma(A) = \mu(A)\}$.

Since $X \in \Pi$, $\Lambda = \mathcal{P}(X)$ (as in the "proof") is a Λ -system in view of (i), (ii)' and (iii) in the proof of Prop 5.2.

Since $\Pi \subseteq \Lambda$, by Dynkin's Thm we have $\sigma(\Pi) \subseteq \Lambda$. ■

Remark 5.7

- (i) The intersection of an arbitrary family of λ -systems is a λ -system. Given $\mathcal{C} \subseteq \mathcal{P}(X)$, $\lambda(\mathcal{C})$ is the λ -system generated by $\mathcal{C}$.
- (ii) If Λ is both a λ -system and a Π -system then it is a σ -algebra.

Indeed, by Def 5.4 (i) + (ii)

$$\phi = X \setminus X \in \Lambda.$$

Also by (i) + (ii), if $A \in \Lambda \Rightarrow X \setminus A \in \Lambda$. (5.1)

Finally, if $\{A_n\} \subseteq \Lambda$, then

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} A'_n, \quad A'_n \subseteq A'_{n+1},$$

$$A'_n = \bigcup_{k=1}^n A_k \in \Lambda \text{ because}$$

$$X \setminus A'_n = \bigcap_{k=1}^n (X \setminus A_k) \in \Lambda \text{ because it is a } \Pi\text{-system} \quad (+ \text{ induction})$$

$\in \Lambda \text{ by (5.1)}$

Hence, again by (5.1) we have that $A'_n \in \Lambda$, and

by Def 5.4 (iii) we conclude that

$$\bigcup_{n=1}^{\infty} A'_n \in \Lambda.$$

end lect #8

Proof of Thm 5.5 [Dynkin Thm]

N.B. σ -algebras are λ -systems
 $\Rightarrow \lambda(\Pi) \subseteq \sigma(\Pi)$ always!

claim 1 $\sigma(\Pi) \subseteq \lambda(\Pi) \subseteq \Lambda$

The last inclusion is trivial since $\Pi \subseteq \Lambda$. All that

So

$\lambda(\pi)$ is the smallest λ -system that contains π .

claim 2 $\lambda(\pi)$ is a σ -algebra.

If so, since $\pi \subseteq \lambda(\pi) \Rightarrow \sigma(\pi) \subseteq \lambda(\pi)$, proving claim 1.

Now by Remark 5.7 (ii), if $\lambda(\pi)$ is a π -system then it is a σ -algebra. So we are "reduced" to proving that $\lambda(\pi)$ is a π -system.

Def 5.3 (i) * By definition of λ -system, $X \in \lambda(\pi)$ hence $\lambda(\pi) \neq \emptyset$ (see Def 5.3).

let $C \in \lambda(\pi)$ and let $\Lambda_C := \{A \in \lambda(\pi) : A \cap C \in \lambda(\pi)\}$.

Note that $\lambda(\pi)$ is a π -system (if)

claim 3 $\Lambda_C = \lambda(\pi)$.

because then given $C \in \lambda(\pi)$ and $A \in \lambda(\pi) \Rightarrow A \cap C \in \lambda(\pi)$

clearly, $\Lambda_C \subseteq \lambda(\pi)$. Conversely, (if) Λ_C is a λ -system $\xrightarrow{\text{claim 4}}$ $\pi \subseteq \Lambda_C \xrightarrow{\text{claim 5}}$

then $\lambda(\pi) \subseteq \Lambda_C$, proving claim 3.

claim 4 Λ_C is a λ -system

(i) $X \in \lambda(\pi)$, and $X \cap C = C \in \lambda(\pi)$, so $X \in \Lambda_C$.

(ii) let $A, B \in \Lambda_C$, $A \subseteq B$.

Then $(B \setminus A) \cap C = (B \cap C) \setminus (A \cap C) \in \lambda(\pi)$ (it is a λ -system).
 $\underbrace{B \cap C}_{\in \lambda(\pi)} \setminus \underbrace{A \cap C}_{\in \lambda(\pi)}$

(iii) let $A_n \in \Lambda_C$, $A_n \subseteq A_{n+1}$, $n \in \mathbb{N}$.

Then

$(\bigcup_{n=1}^{\infty} A_n) \cap C = \bigcup_{n=1}^{\infty} (A_n \cap C) \in \lambda(\pi)$ (it is a λ -system)
 $\underbrace{\quad}_{\in \lambda(\pi) \text{ and increasing}}$

claim 5 $\Pi \subseteq \Lambda_c$

let $A \in \Pi$. we show that $A \cap C \in \lambda(\Pi)$.

If $C \in \Pi$, since Π is a Π -system then, $A \cap C \in \Pi \subseteq \lambda(\Pi)$.

hence $\Pi \subseteq \Lambda_c$. (*)

If $C \in \lambda(\Pi)$ (and not, necessarily, belongs to Π), then by the previous case, because Π is a Π -system

$$\Pi \subseteq \Lambda_A \text{ (see (*))}$$

and by claim 4, Λ_A is a λ -system, so

$$\lambda(\Pi) \subseteq \Lambda_A.$$

In particular, $C \in \Lambda_A \Rightarrow C \cap A \in \lambda(\Pi) \Rightarrow A \in \Lambda_c$. ■

6. REGULARITY

Regularity properties of measures concern the possibility of approximating in measure one class of sets, e.g., nonmeasurable sets, by another class of sets, e.g., measurable sets.

As we'll see, Lebesgue measure is Borel regular in the sense that Lebesgue measurable sets can be approximated in measure from outside by open sets and from the inside by closed sets. Further, they can be approximated by Borel sets up to sets of measure zero.

(Def 3.5) Def 6.1 An outer measure $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ is said to be regular if for all $E \subseteq X$ there exists a μ^* -measurable set $F \subseteq X$ such that $E \subseteq F$ and $\mu^*(E) = \mu^*(F)$.

Prop 6.2 let $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ be a regular outer measure.

If $\{E_n\} \subseteq \mathcal{P}(X)$ is an increasing sequence, then

$$\mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu^*(E_n).$$

N.B. In Prop 4.6 we proved this result for measures.

Proof Since μ^* is regular, for all $n \in \mathbb{N}$ let F_n be μ^* -measurable with $E_n \subseteq F_n$, $\mu^*(E_n) = \mu^*(F_n)$. If $\exists n \in \mathbb{N}: \mu^*(E_n) = +\infty$ then otherwise, By Carathéodory Thm 4.9, the sets $G_n := \bigcap_{k=n}^{\infty} F_k$ are μ^* -measurable, with $E_n \subseteq G_n \subseteq F_n$, and $\mu^*(G_n) \leq \mu^*(F_n) = \mu^*(E_n) \leq \mu^*(G_n) \Rightarrow \mu^*(G_n) = \mu^*(E_n)$. Since $G_n \subseteq G_{n+1}$, and as $\mu^*: \mathcal{M}^* \rightarrow [0, +\infty]$ is a measure (Thm 4.9), by Prop 4.6 we have

$$\begin{aligned} \mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) &\leq \mu^*\left(\bigcup_{n=1}^{\infty} G_n\right) = \lim_{n \rightarrow \infty} \mu^*(G_n) = \\ &= \lim_{n \rightarrow \infty} \mu^*(E_n) \leq \mu^*\left(\bigcup_{n=1}^{\infty} E_n\right), \end{aligned}$$

and the result follows. (monotonicity)

Remark 6.3 This proposition may fail if μ^* is not regular.

let $X := \mathbb{N}$, and define

$$\mu^*(E) := \begin{cases} 0 & \text{if } E = \emptyset, \\ 1 & \text{if } E \text{ is finite,} \\ 2 & \text{if } E \text{ is infinite,} \end{cases} \quad E \subseteq \mathbb{N}.$$

Then μ^* is an outer measure, and if $E_n = \{1, \dots, n\}$ then

$$2 = \mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) \neq \lim_{n \rightarrow \infty} \mu^*(E_n) = 1.$$

Remark 6.4 Given an outer measure $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$, we can construct a regular outer measure by defining

$$\nu^*(E) := \inf \{ \mu^*(F) : E \subset F, F \text{ is } \mu^*\text{-measurable} \}$$

It can be shown that

- (i) E is μ^* -measurable $\Rightarrow E$ is ν^* -measurable and $\mu^*(E) = \nu^*(E)$.
- (ii) E is ν^* -measurable and $\nu^*(E) < +\infty \Rightarrow E$ is μ^* -measurable.

Def 6.5 Let X be a topological space, and let $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ be an outer measure.

- (i) A set $E \subset X$ is said to be inner regular if

$$\mu^*(E) = \sup \{ \mu^*(K) : K \subset E, K \text{ compact} \},$$

and is outer regular if

$$\mu^*(E) = \inf \{ \mu^*(U) : E \subset U, U \text{ open} \}.$$

- (ii) a set $E \subset X$ is said to be regular if it is both inner and outer regular.

Def 6.1 (VS) Def 6.5 N.B. Do not confuse "a set being regular" with "a measure being regular."

Def 6.6 Let X be a topological space, and let $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ be an outer measure.

- (i) μ^* is said to be a Borel outer measure if

$\mathcal{B}(X) \subseteq \mathcal{M}^{\mu^*}$, i.e., every Borel set is μ^* -measurable;

- (ii) μ^* is said to be a Borel regular outer measure

if μ^* is a Borel outer measure

- for all $E \subset X$ there exists a Borel set $B \subset X$ such that $E \subset B$ and $\mu^*(E) = \mu^*(B)$.

Def 6.7 let X be a topological space,
and let $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ be an outer measure.

μ^* is said to be a Radon outer measure if

(i) μ^* is a Borel outer measure;

(ii) $\mu^*(K) < +\infty$ for every compact set $K \subset X$;

(iii) every open set $A \subset X$ is inner regular;

(iv) every set $E \subset X$ is outer regular.

[So every open set is regular, see Def 6.5 (ii)]

Remark 6.8 A Radon outer measure is Borel regular.

Indeed, let $E \subset X$. If $\mu^*(E) = +\infty$ then note that X is open,
 $X \supset E$, and $\mu^*(E) \leq \mu^*(X) = +\infty$.

If $\mu^*(E) < +\infty$, by $\overset{(\text{iii})}{\text{outer regularity}}$ for all $n \in \mathbb{N}$ we
may find open set U_n , $E \subset U_n$,

$$\mu^*(U_n) \leq \mu^*(E) + \frac{1}{n}.$$

Then the Borel set

$$B := \bigcap_{n=1}^{\infty} U_n$$

contains E , and for all $n \in \mathbb{N}$,

$$\mu^*(E) \leq \mu^*(B) \leq \mu^*(U_n) \leq \mu^*(E) + \frac{1}{n} \rightarrow \mu^*(E),$$

and so,

$$\mu^*(E) = \mu^*(B).$$

The converse is false in general.

Example 6.9 let X be an uncountable set with the
discrete topology $\mathcal{C} = \mathcal{P}(X)$. For $E \subset X$ define

$$\mu(E) = \begin{cases} 0 & \text{if } E \text{ is countable,} \\ \infty & \text{otherwise.} \end{cases}$$

Then $\mu: \mathcal{P}(X) \rightarrow [0, +\infty]$ is a measure.

Since every subset of X is open, μ is Borel regular and every set is outer regular.

Compact sets have to be finite, so $K \subset X$ compact $\Rightarrow \mu(K) = 0$.
Hence, uncountable sets cannot be inner regular, and μ is not a Radon measure.

Why: $\{U \in \mathcal{O} : U \supset K\}$ is an open cover of K that does not admit a finite subcover unless K is finite.

Still, a partial result toward Borel regular $\Rightarrow$ Radon is:

Proposition 6.10 Let X be a locally compact Hausdorff space such that every open set $A \subseteq X$ is σ -compact (i.e., $A = \bigcup_{n=1}^{\infty} K_n$, K_n compact). Let $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ be a Borel outer measure such that $\mu^*(K) < \infty$ for every compact set $K \subseteq X$.

Then every Borel set is inner regular and outer regular.

If, in addition, μ^* is a Borel regular outer measure, then it is a Radon outer measure.

[see also Thm 6.2]

Proof Since X is σ -compact, we may write

$$X = \bigcup_{n=1}^{\infty} K_n, \quad K_n \subseteq K_{n+1}, \quad K_n \text{ compact.}$$

Because X is a locally compact Hausdorff space, $\forall n \in \mathbb{N}$ we may find U_n open such that $K_n \subset U_n$ and $\overline{U_n}$ is compact. Wlog, assume that $U_n \subseteq U_{n+1}$, $n \in \mathbb{N}$.

Step 1 Fix $n \in \mathbb{N}$. We prove that every Borel subset of U_n is inner and outer regular w.r.t. outer measure $\mu^* \llcorner U_n$ (restriction of μ^* to U_n): $\beta(U_n) \rightarrow [0, +\infty]$ with $\mu^* \llcorner U_n(E) := \mu^*(E)$, $E \subseteq U_n$.

Since $\bar{U}_n$ is compact, $\mu^* \llcorner U_n$ is a finite outer measure.
let $\mathcal{M} := \{E \subset U_n : E \text{ is } \mu^* \llcorner U_n \text{ inner and outer regular}\}$.

This step will be accomplished if we show that

(i) $\mathcal{M}$ contains all open subsets of U_n ;

(ii) $\mathcal{M}$ is a σ -algebra.

(i) If $V \subset U_n$ is open, then clearly it is outer regular.

Since V is G -compact, there exists an increasing sequence of compact sets $C_j \subset V$ with

$$V = \bigcup_{j=1}^{\infty} C_j$$

and so, since $V, C_j \in \beta(X) \subseteq \mathcal{M}^*$ (i.e.,

(because μ^* is a Borel outer meas.) V, C_j and μ^* -measurable), in view of Carathéodory

Thm 4.9 and by Thm 4.6 we have

$$\mu^*(V) = \mu^*\left(\bigcup_{j=1}^{\infty} C_j\right) = \lim_{j \rightarrow +\infty} \mu^*(C_j).$$

We conclude that V is inner regular.

(ii) clearly $\emptyset \in \mathcal{M}$, and by (i) $U_n \in \mathcal{M}$.

claim $\mathcal{M}$ is closed w.r.t. countable unions.